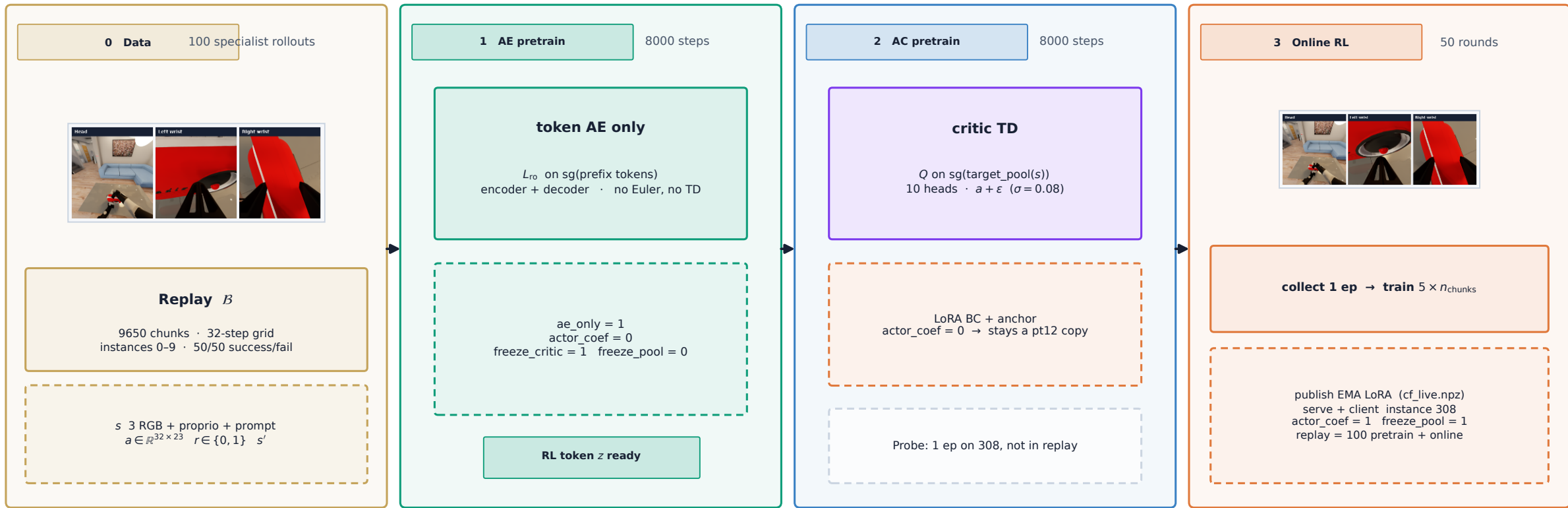




Joint loss (every stage; λ_π gates only $-Q$)

$$L = 4 L_{td} + \lambda_\pi (-\text{mean} \min_k Q_k(s, a)) + \beta \|a - \tilde{a}\|^2 + L_{bc} + L_{ro}$$

$\lambda_\pi = 0$ in AE / AC pretrain, 1 online. Anchor $\beta = 100$ and BC stay on. Kill if $|Q| > 2$ or actor-ref RMSE > 0.04 .